

Executive Summary – Telco Customer Churn Analysis

This project analyzes customer churn behavior in a telecom company using data visualization and exploratory data analysis techniques. The overall churn rate is **26.54%**, indicating that approximately **1 out of every 4 customers discontinues the service**, which highlights a critical need for retention strategies.

The analysis identifies **customer tenure as a major churn driver**. Customers with a tenure of **0–2 months exhibit the highest churn rates (around 45–50%)**, while those with tenure beyond **24 months show significantly lower churn (below 15%)**. This indicates that **customer retention efforts should be concentrated during the early lifecycle stage**.

Contract type plays a crucial role in churn behavior. Customers on **month-to-month contracts contribute to the highest churn segment (~40–45%)**, whereas **one-year contracts reduce churn to ~10–15%**, and **two-year contracts further reduce churn to below 5%**. This demonstrates that **long-term commitments strongly improve customer retention**.

Service adoption patterns reveal that **customers without additional services are more likely to churn**:

- **Online Security, Tech Support, and Device Protection** absence leads to churn rates of **~35–45%**
- Customers who subscribe to these services show significantly lower churn (**~15–20%**)

This suggests that **bundling and promoting value-added services can effectively reduce churn**.

From a product perspective, **Fiber Optic internet users show higher churn (~40%) compared to DSL users (~20%)**, indicating potential concerns related to **pricing, service quality, or customer expectations**.

Payment method analysis highlights behavioral trends:

- Customers using **Electronic Check** exhibit the highest churn (**~45%**)
- Customers using **Auto-pay methods (credit card/bank transfer)** show lower churn (**~15–20%**)

This implies that **frictionless and automated payment systems contribute to better retention**.

Additionally, **higher monthly charges (>70 units)** are associated with increased churn (**~35–40%**), especially among customers with short tenure and flexible contracts. In contrast, customers with **lower monthly charges show reduced churn (~15–20%)**, reflecting **price sensitivity as a key factor**.

Demographic analysis indicates that **senior citizens have a higher churn rate (~40%) compared to non-seniors (~25%)**, suggesting the need for **better support systems and personalized engagement strategies** for this segment.

Key Business Insights

- **26.54% overall churn rate** indicates moderate-to-high customer loss
 - **Early-stage customers (0–2 months) are most vulnerable (~45–50% churn)**
 - **Month-to-month contracts drive maximum churn (~40–45%)**
 - **Lack of value-added services increases churn up to ~45%**
 - **Fiber optic users and electronic check users show highest churn (~40–45%)**
 - **High monthly charges correlate with higher churn (~35–40%)**
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Conclusion

Customer churn in the telecom sector is driven by a combination of **short-term contracts, low service engagement, high pricing, and suboptimal customer experience**. To reduce churn, the company should:

- Strengthen **early customer onboarding and engagement**
- Promote **long-term contract plans**
- Increase adoption of **value-added services**
- Improve **payment experience through automation**
- Optimize **pricing and service quality, especially for fiber users**